

Lie Groups

1.1 Lie Subgroups

Definition 1.1.1

Suppose G is a Lie group. A **Lie subgroup of G** is a subgroup of G endowed with a topology and smooth structure making it into a Lie group and an *immersed submanifold* of G .

Remark.

- A Lie subgroup of a compact Lie group need not be compact.